

SUNDAS SALEEM

SENIOR SOFTWARE ENGINEER

Backend Systems, Distributed Services & AI Infrastructure

Building scalable, reliable platforms that improve performance and engineering efficiency.

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Senior Software Engineer with 6+ years of experience architecting scalable backend platforms, distributed systems, and AI-driven automation solutions for high-volume, mission-critical environments. Expertise in Python, C++, cloud-native technologies, microservices, and event-driven architectures, delivering resilient systems processing 2M+ daily interactions with zero downtime. Proven track record of improving system reliability, observability, and performance through large-scale platform modernization, automation, and optimization initiatives. Skilled in building real-time analytics, intelligent orchestration, and telephony solutions that drive operational efficiency and business impact. Recognized for technical leadership, complex problem-solving, and delivering high-performance software solutions at scale.

Skills & Expertise

Backend & Systems Architecture: Backend Software Development, Distributed Systems Architecture, Microservices Development, Event-Driven Architecture, API Design & Integration.

Programming & AI Development: Python Development, C++ Development, AI Agent Development, LLM Integration.

Cloud & Data: Cloud & AWS Engineering, Real-Time Data Processing, Database Design & Optimization.

Quality & Operations: Observability & Monitoring, Performance Optimization, Test Automation, CI/CD.

Testing & Observability: Test-driven development (TDD), Unit testing (Google Test, Python frameworks), automated regression testing, load testing, Selenium, Cucumber, OpenTelemetry metrics, CI/CD pipelines.

Technical Proficiencies

Programming: Python, C++, C#, Java, JavaScript, Node.js, React, TypeScript, SQL, Bash

Backend & APIs: FastAPI, Django, REST APIs, gRPC, Node.js, WebSocket, Socket.io, TCP/IP

Cloud & DevOps: AWS, Docker, Docker API, Kubernetes, Linux, Git, Jenkins

Databases: PostgreSQL, MySQL, Redis

Data & Observability: Kafka, OpenTelemetry, Prometheus, Grafana, Sentry, Jupyter, Matplotlib

AI & Contact Center: MCP (Model Context Protocol), AI Agent Frameworks, Amazon Connect, Cisco, Five9, Twilio, Avaya, Genesys, SIP/Call Dialing

Professional Experience

Senior Software Engineer | Afiniti, Pakistan

Jan 2025 – Jun 2026

Drove technical leadership for backend architecture, AI-enabled platform development, and cloud-native modernization initiatives supporting enterprise-scale contact center solutions.

- ◆ Reduced critical production incidents by ~90% by redesigning a single-threaded telephony routing service into a scalable multi-threaded architecture capable of processing 2M+ call events while maintaining 100% service availability.
- ◆ Enabled reliable processing of 2M+ calls per day with zero downtime by developing high-performance backend services, fail-safe routing mechanisms, and large-scale load-testing frameworks for telephony platforms.
- ◆ Implemented OpenTelemetry-based observability framework across 10+ Python microservices and architected reusable instrumentation library, reducing troubleshooting effort by 70%+ through distributed tracing and real-time monitoring.
- ◆ Owned the architecture of AI agents and Model Context Protocol (MCP) services, influencing platform modernization while enabling intelligent engineering automation and service orchestration.
- ◆ Built a real-time analytics platform using Kafka and Python while developing FastAPI-based AWS orchestration services, enabling live operational visibility and scalable call-routing automation across distributed environments.
- ◆ Improved database query performance by 40% and reduced database load by 30%+ through query optimization, caching strategies, and backend performance tuning.

- ◆ Demonstrated technical leadership for backend architecture, engineering standards, and CI/CD practices while guiding code reviews, influencing platform direction, and increasing feature adoption by 25%.
- ◆ Owned switch interface services across multiple ACD/CX systems, primarily NICE inContact, maintaining socket-based SDK, REST, and WebSocket integrations to power AI-driven call routing.
- ◆ Led customer migration support between ACD platforms (NICE inContact), analyzing existing API implementations and documenting integrations in OpenAPI format to enable the customer's development team to build middleware efficiently.
- ◆ Implemented PII data protection across switch interface services, ensuring secure handling of sensitive customer data across all integrations.
- ◆ Integrated OpenTelemetry metrics into switch interface services via an internal wrapper library, delivering company-wide observability and enabling SRE teams to monitor service performance and reliability.
- ◆ Leveraged enterprise LLMs (GitHub Copilot in VS Code) and custom MCP servers to accelerate feature delivery, raise unit test coverage, streamline code review, and speed up log analysis and bug resolution — improving overall code quality and engineering velocity.

Software Engineer | Afiniti, Pakistan

Oct 2021 – Jan 2025

Owned the design and delivery of scalable backend services, telephony integrations, and platform enhancements, partnering across engineering teams to improve product reliability and operational performance.

- ◆ Reduced manual testing effort by 50% and accelerated release cycles by 20% through the development of automated testing frameworks using C++, Python, C#, NodeJS and JavaScript.
- ◆ Directed architectural evolution of a mission-critical .NET contact center platform, mentoring teammates on platform capabilities while ensuring uninterrupted feature delivery.
- ◆ Improved database response times from minutes to seconds by identifying performance bottlenecks, implementing indexing strategies, and optimizing complex MySQL queries.
- ◆ Enabled large-scale production validation by implementing Amazon Connect load-testing frameworks capable of simulating thousands of agents and millions of call interactions.
- ◆ Cut troubleshooting time from hours to minutes and improved incident resolution efficiency by 45% through Python-based log analytics platform that automated diagnostics, visualization, and analysis of production telephony workflows.
- ◆ Decreased integration defects by 40% through the design of robust REST APIs and integration of Twilio, Five9, Avaya, and NICE InContact platforms, enabling scalable real-time communication workflows.
- ◆ Influenced the technical roadmap for mission-critical backend enhancements, designing scalable data architectures while driving cross-functional delivery and achieving a 98% delivery rate.

Analyst Software Engineer | Afiniti, Pakistan

Dec 2019 – Oct 2021

Developed automation frameworks and platform validation solutions that improved release quality, system scalability, and engineering productivity across distributed telecom platforms.

- ◆ Built a distributed C++ integration testing platform covering ~90% of microservices, significantly improving release confidence and enabling automated regression testing across the product ecosystem.
- ◆ Simulated millions to billions of call and agent interactions through a high-scale performance validation framework, while designing multi-threaded queue management and batch-processing capabilities for production-grade load testing.
- ◆ Reduced manual validation efforts by 70% and enabled early detection of service crashes, database issues, and performance bottlenecks through automated testing, monitoring, and CI/CD-integrated validation frameworks.
- ◆ Designed REST APIs, dynamic configuration capabilities, and reusable validation components that streamlined test execution, scenario management, and platform testing activities.
- ◆ Created a Node.js-based simulation platform supporting REST, TCP, and WebSocket protocols, enabling realistic end-to-end testing and validation of distributed communication workflows.
- ◆ Improved system reliability and integration quality by automating validation of asynchronous event flows across backend services, telecom platforms, and distributed microservices.
- ◆ Developed reusable testing frameworks adopted as engineering standards while contributing to platform architecture decisions, scalability initiatives, and long-term product roadmap planning.

- ◆ Partnered with senior engineers to shape automation architecture, establish reusable engineering frameworks, and improve long-term platform scalability and release quality.

Education

Bachelor of Science in Software Engineering | NED University of Engineering & Technology, PK